ASSIGNMENT-10.4

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B-16

Task 1 – Fix Python Errors:-

Code:-

```
def calc(a,b)
print("sum:",a+b)
print("product",a*b)

calc(5 3)
```

```
  File "/tmp/ipykernel_234/3581734860.py", line 1
    def calc(a,b)
                ^
SyntaxError: expected ':'
```

Next steps: Explain error

Prompt:-

Review this Python code and fix syntax errors, indentation issues,incorrect variable names and faulty function calls.Explain each correction.

Correct Code:-

```
def calc(a, b):
    print("Sum:", a + b)
    print("Product:", a * b)

calc(5, 3)
```

```
Sum: 8
Product: 15
```

Task 2 – Performance Optimization

Code:-

```python
def find_duplicates(arr):
    duplicates = []
    for i in range(len(arr)):
        for j in range(i+1, len(arr)):
            if arr[i] == arr[j]:
                duplicates.append(arr[i])
    return duplicates

print(find_duplicates([1,2,3,2,4,5,1]))
```

```
[1, 2]
```

Prompt:-

Analyze this Python code and optimize it for performance.Explain why the current solution is inefficient and improvetime complexity.

Correct Code:-

```python
def find_duplicates(arr):
    seen = set()
    duplicates = set()

    for num in arr:
        if num in seen:
            duplicates.add(num)
        else:
            seen.add(num)

    return list(duplicates)

print(find_duplicates([1,2,3,2,4,5,1]))
```

```
[1, 2]
```

Task 3 – Improve Readability

Code:-

```python
def f(x,y):
 r=x*y
 if r>100:print("big")
 else:print("small")
 return r
```

Prompt:-

Refactor this Python code to improve readability,apply PEP8 formatting,use better variable names and add documentation.

Correct Code:-

```python
def calculate_product(num1, num2):
    """
    Calculates the product of two numbers and
    checks if the result is large or small.
    """

    result = num1 * num2

    if result > 100:
        print("Result is large")
    else:
        print("Result is small")

    return result
```

Task 4 – Secure Coding

Code:-

```python
import sqlite3

def get_user(username):
    conn = sqlite3.connect("users.db")
    query = "SELECT * FROM users WHERE username = '" + username + "'"
    result = conn.execute(query)
    return result.fetchall()
```

Prompt:-

Identify security vulnerabilities in this code.Rewrite it using secure coding practices,parameterized SQL queries, input validation,and proper exception handling.

Correct Code:-

```python
import sqlite3

def get_user(username):
    try:
        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()

        query = "SELECT * FROM users WHERE username = ?"
        cursor.execute(query, (username,))

        result = cursor.fetchall()
        return result

    except sqlite3.Error as error:
        print("Database error:", error)

    finally:
        conn.close()
```

Task 5 – AI Code Review Report

Code:-

```python
def c(a,b):print(a+b)
```

Prompt:-

Act as an AI code reviewer.Analyze the following Python code for readability,naming conventions, formatting, documentation,maintainability and error handling.Generate a structured code review report.

Correct Code:-

```python
def calculate_sum(num1, num2):
    """Returns the sum of two numbers."""
    return num1 + num2
```